				[2]			
17	**(i)**		$\frac{A}{(x+1)}+\frac{B}{(x-2)}+\frac{C}{(x-2)^2}$	**B1**	**3.1a**	may be seen later	
			$x^2-8x+9=A(x-2)^2+B(x+1)(x-2)+C(x+1)$	**M1**	**2.1**		
			$A=2$	**A1**	**1.1**		
			$B=-1$	**A1**	**1.1**	$\frac{2}{(x+1)}-\frac{1}{(x-2)}-\frac{1}{(x-2)^2}$	
			$C=-1$	**A1**	**1.1**		
				[5]			

Question			Answer	Marks	AOs	Guidance	
17	(ii)		$\int \frac{dy}{y} = \int \frac{x^2 - 8x + 9}{(x+1)(x-2)^2} dx$ soi	**M1***	**3.1a**	allow omission of integral signs and/or omission of dy and/or dx	
			use of their partial fractions in integration	**M1***	**2.1**	allow one sign error and/or one coefficient error	condone use of brackets instead of modulus signs; these two **A** marks are only available following the award of **both M** marks
			$\ln\lvert y\rvert = 2\ln\lvert x+1\rvert - \ln\lvert x-2\rvert + \frac{1}{x-2} + c$	**A1**	**1.1**	**A1** for any correct natural log integral on RHS FT *their* $\frac{2}{x+1}$ or *their* $\frac{-1}{x-2}$	
				A1	**1.1**	**A1** for $\frac{1}{x-2}$ FT *their* $\frac{k}{(x-2)^2}$	
			substitution of $y = 16$ and $x = 3$	**M1dep***	**1.1**	expression must include + c and must include at least one natural log term; may be awarded after exponentiating	may be awarded following collection of like terms, which may contain errors NB $c = -1$
			correctly exponentiate both sides of their equation	**M1**	**1.1**		
			$y = \frac{(x+1)^2}{x-2} e^{\frac{3-x}{x-2}}$ oe	**A1**	**2.1**	eg $\frac{(x+1)^2}{x-2} e^{\frac{1}{x-2}} e^{-1}$	
				[7]			